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Master Asset Provenance & Architectural Genesis

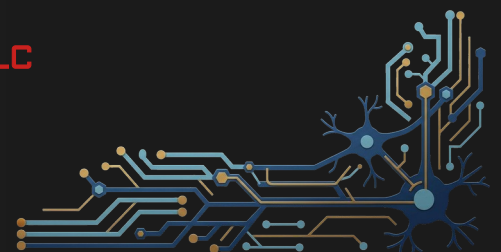
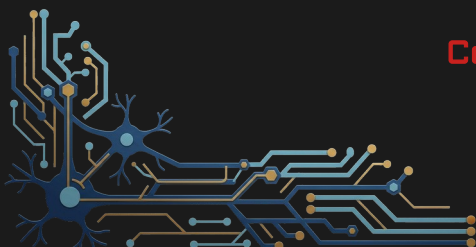
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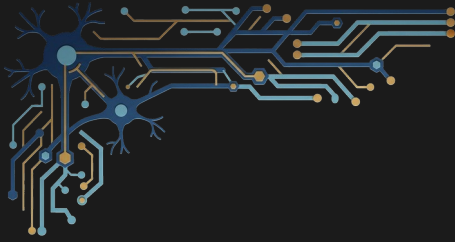
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DOCUMENT ID: NBPR-2026-0025

SUBJECT: Master Asset Provenance & Architectural Genesis

PRINCIPAL ARCHITECT: Kenneth Renshaw (CEO & CSO)

ENTITY: Neurobotics LLC | Yale, Oklahoma

CLASSIFICATION: Executive Data Room Vault / Restricted IP Verification

Master Asset Provenance & Architectural Genesis

[1.0] Executive Profile & Engineering Pedigree

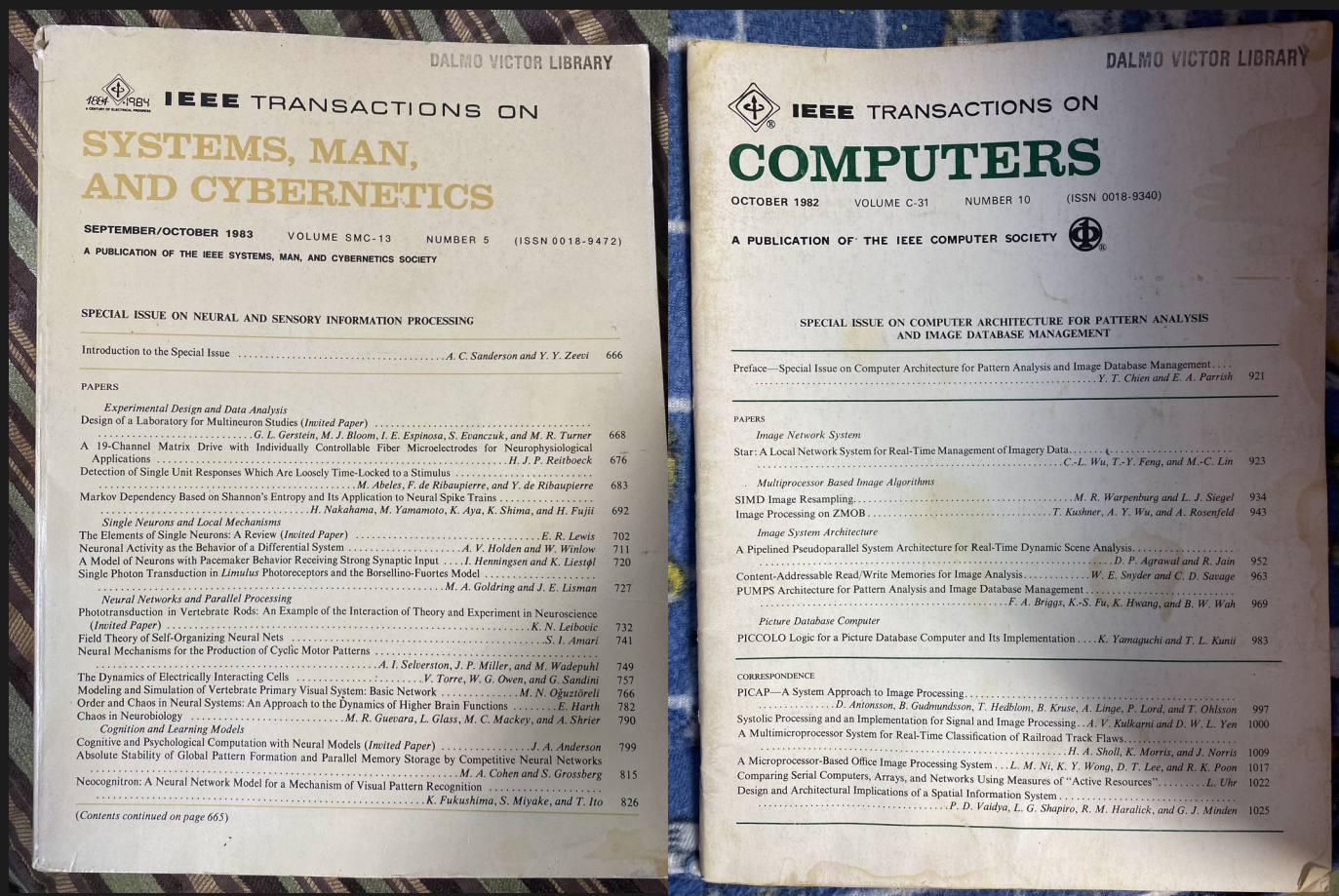
Kenneth Renshaw is the Co-Founder, Chief Executive Officer, and Chief Science Officer of Neurobotics LLC. With over 50 years of continuous bare-metal computation, hardware design, theoretical physics, and neuroanatomical modeling, he is the sole primary architect and inventor of the Silicon-Based Artificial Life Form (SBALF) and Silicon-Based Utility Droid (SBUD) architecture.

His engineering instinct was forged in zero-tolerance aerospace, electronic warfare, and deep-space mission environments where operational failure was not an option:

- **Aerospace Electronic Warfare (Dalmo Victor / Bell Aerospace):** Designed radar warning systems for the F-14 Tomcat and EF-111A Raven. Physical library artifacts (such as stamped 1982 IEEE Transactions on Computers journals from the Dalmo Victor Library) confirm his early

professional immersion in real-time pattern analysis and parallel image processing architectures.

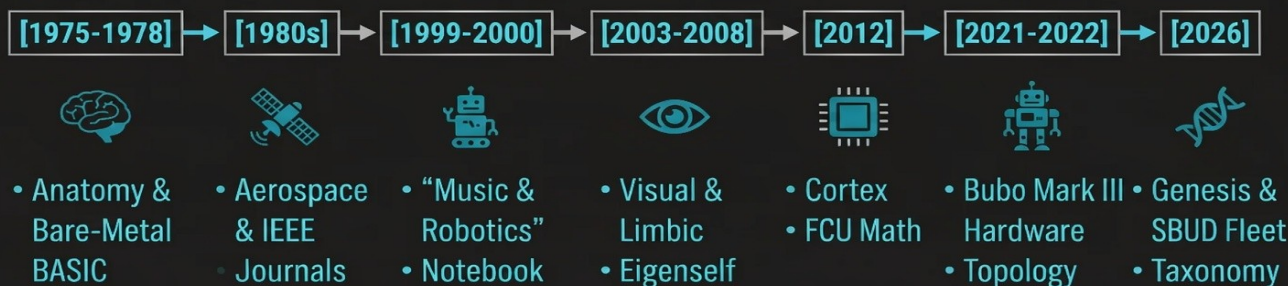
- **Deep-Space Computing (Litton Industries Applied Technology):** Engineered ATAC-16 spacecraft computers for the Galileo Space Probe. This work utilized bit-slice architecture—cascading 4-bit ALUs to build custom 16-bit spaceborne processors, authoring custom microcode, and designing radiation-hardened hardware with voter circuits and error correction for survival in deep space.
- **Global Air Safety (TCAS II):** Key contributor to the Traffic Alert and Collision Avoidance System (TCAS II), establishing the foundational design philosophy that governs all Neurobotics systems: "Design so that the failure mode is not death."
- **Large-Scale Systems Infrastructure:** Decades of leadership in distributed systems, bare-metal hardware, and enterprise computing at Sun Microsystems (SPARCstation era), Oracle, Yahoo!, SGI, CivicActions, and Teleflora.



September/October 1983 *IEEE Transactions of Systems, Man, and Cybernetics* | October 1982 *IEEE Transactions on Computers* with the Official **Dalmo Victor Library** stamp; From the **Personal Library of Kenneth Renshaw**

[2.0] Complete Chronological Prior Art Timeline

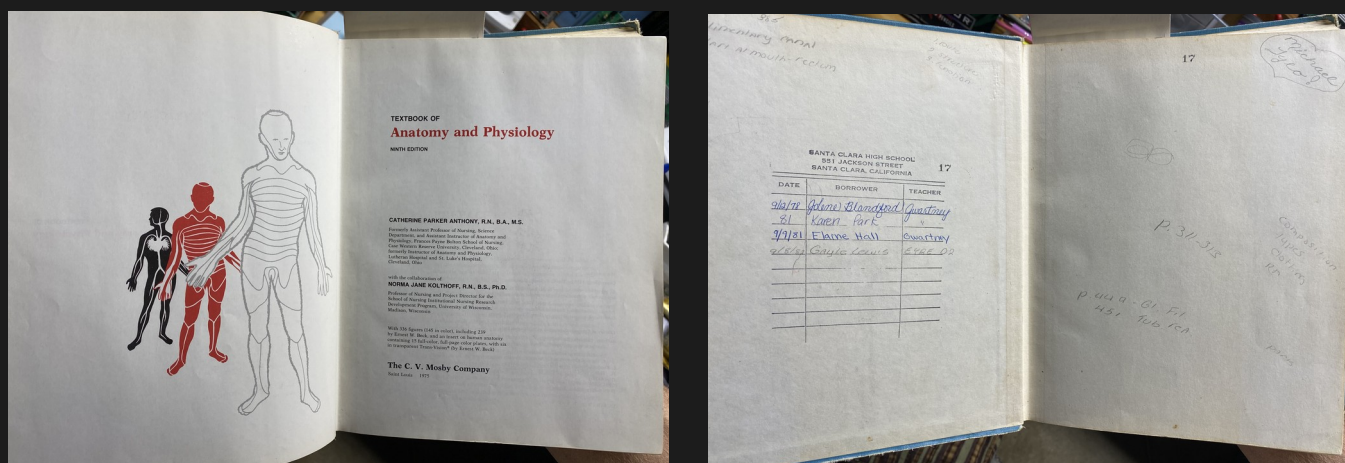
THE EVOLUTION OF NEUROBOTICS: COMPLETE PROJECT TIMELINE



The SBALF architecture is not an adaptation of recent large language models or cloud-based software trends; it is the culmination of a half-century of continuous physical ledgers, handwritten notebooks, and empirical biological modeling.

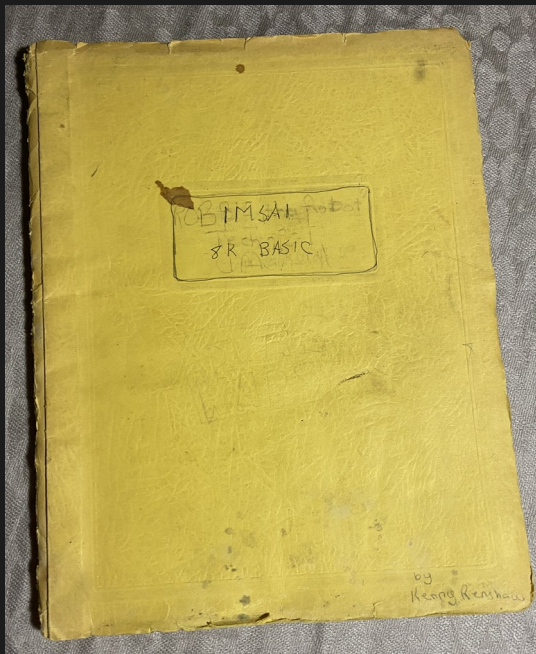
1. 1975–1978: Anatomical Bedrock & Bare-Metal Foundations

- **Anatomy & Physiology Ledger:** A 1975 Textbook of Anatomy and Physiology (9th Ed., Mosby), checked out at Santa Clara High School in 1978, contains the exact Brodmann area maps, sensory receptor pathways, and spinal interneuron layouts studied and taught by Kenneth while serving as class tutor at age 16.

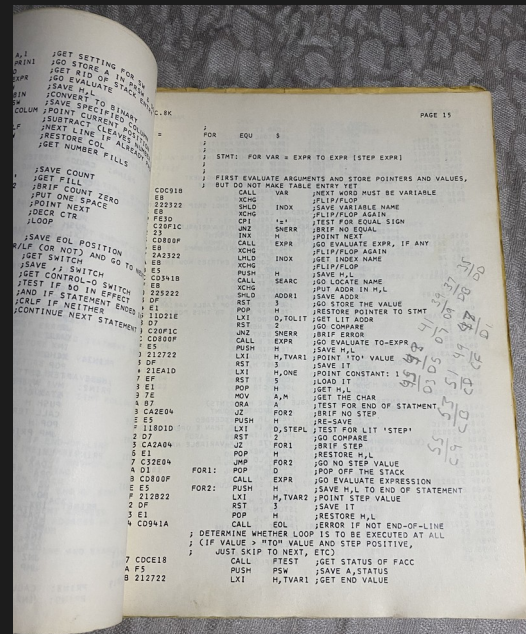


The Inside cover page of the 1975 *Textbook of Anatomy and Physiology Ninth Edition* by The C.V. Mosby Co. from Santa Clara High School | Santa Clara High School checkout card inside the Mosby Anatomy book; From Kenneth Renshaw's Private Library

- **1977 IMSAI 8K Assembly Folder:** A bound repurposed yellow cardboard 1970's three-brad binder signed "Kenny Renshaw" containing hand-annotated Intel 8080/Z80 assembly code (STFAC, LDFAC, floating-point accumulators, CP/M memory equates) for an IMSAI 8K BASIC interpreter. Originally titled "ROBBIE the Robot Technical Manual" (erased and re-captioned "Schematics"), including a previously erased and barely legible "STAR WARS" hand-written title text from the original release of *Star Wars IV: A New Hope* below the notebook title area, this establishes a 50-year prior art claim for hyper-efficient, register-level compute.



Cover of hand-annotated 1977 IMSAI 8K Assembly Code "Robbie the Robot Schematics – STAR WARS" repurposed yellow notebook; From Kenneth Renshaw's Private Library



Inside the hand-annotated 1977 IMSAI 8K Assembly code repurposed yellow notebook; From Kenneth Renshaw's Private Library

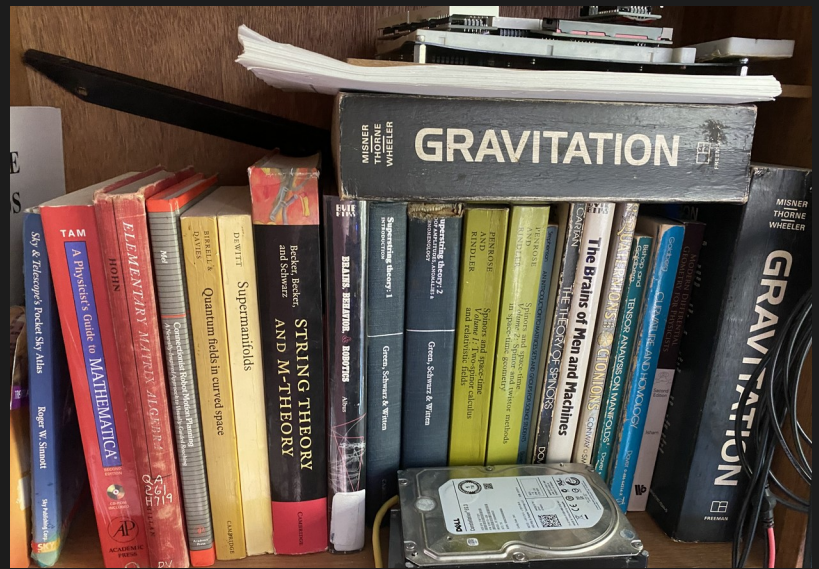
2. 1980s: Pattern Processing & Cybernetic Theory

- **Dalmo Victor IEEE Research:** Dated IEEE publications (October 1982 & September / October 1983) stamped by the Dalmo Victor Library detail pipeline pseudo-parallel processing, SIMD image resampling, and real-time scene analysis architectures.

- **Early Neural Modeling:** Direct implementation of early cybernetic literature (Ernest Kent's *The Brains of Men and Machines*, Albus's *Brains, Behavior, & Robotics*, and Heiserman's autonomous pet schematics).



A shelf of vintage *BYTE* and *Robotics Age* magazines from the 1980's era plus miscellaneous books and notebooks; From Kenneth Renshaw's Private Library

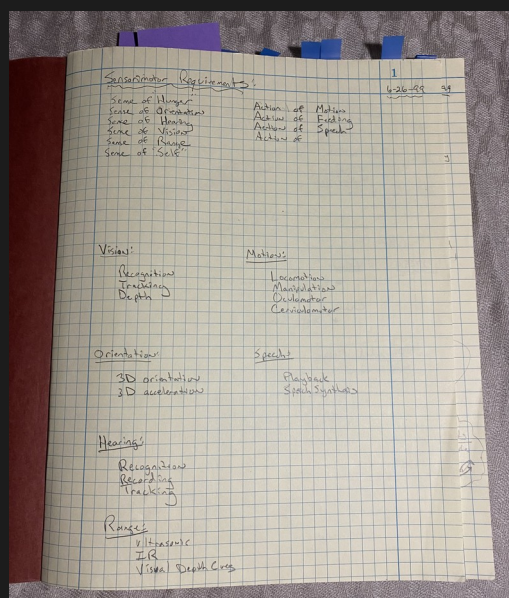


A shelf of vintage robotics and physics books from the 1980's era, plus a well-worn original copy of James Albus' *The Brains of Men and Machines*; From Kenneth Renshaw's Private Library

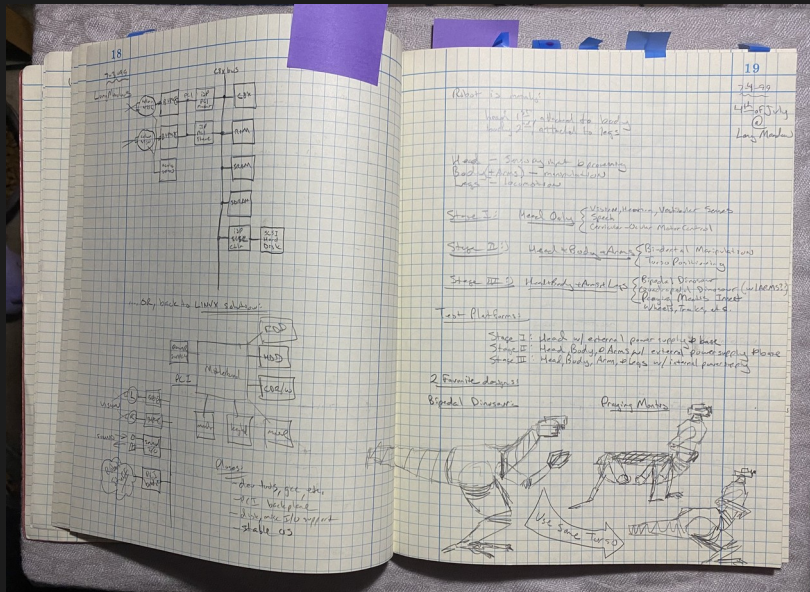
3. 1999–2000: The Sensorimotor "Music & Robotics" Ledgers

- **Bound Computation Notebooks (June-July 1999):** Densely filled grid notebooks detailing sensorimotor requirements ("Sense of Hunger," "Sense of Self," "Binocular Foveated Vision"), dual CMOS image processing, and 4-tier robotic test platforms (bipedal and quadrupedal chassis).

"Music and Robots" brown computational grid-paper notebook grid page with the exact handwritten list of sensorimotor requirements; From Kenneth Renshaw's personal library.



- **DSP Hardware Pipelines (March 2000):** Comprehensive schematic layouts utilizing Analog Devices SHARC and Texas Instruments C6701 DSP chips to route real-time audio, vestibular balance, and foveal vision over local bus topologies.

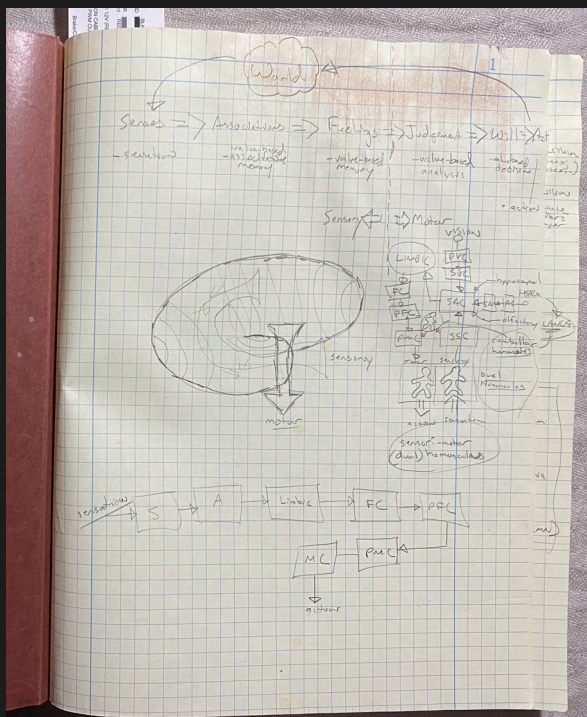


"Music and Robots" brown computational grid-paper notebook grid pages with the handwritten design notes and sketches for a Bipedal/Mantid; From Kenneth Renshaw's personal library.

4. 2003–2008: Visual Hierarchy, Limbic Eigenself, & Biomechanics

- **April/May 2003 "Red Notebook":** Explicit mapping of the Basal Ganglia, Lower Motor Neurons (LMN), and Cerebellar Layers (Purkinje cells, Inferior Olive), establishing the genesis of the S-M-R pipeline:

World -> Senses -> Associations -> Feelings -> Judgment -> Will -> Act



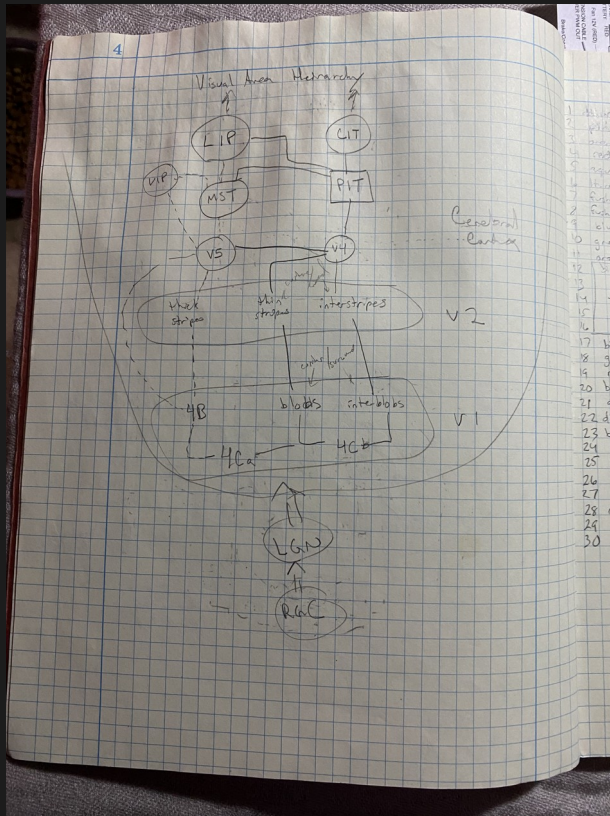
April/May 2003 "Red Notebook" (S-M-R Genesis) brown computational grid-paper notebook grid pages with the handwritten neural biomechanics design notes and sketches; From Kenneth Renshaw's personal library.

- **May 2003 Notebooks:** Complete, color-coded mapping of all 52 Brodmann areas, establishing the biological pipeline:

RGC -> LGN -> V1 -> V2 -> V4/V5 -> PIT/CIT/LIP/MST

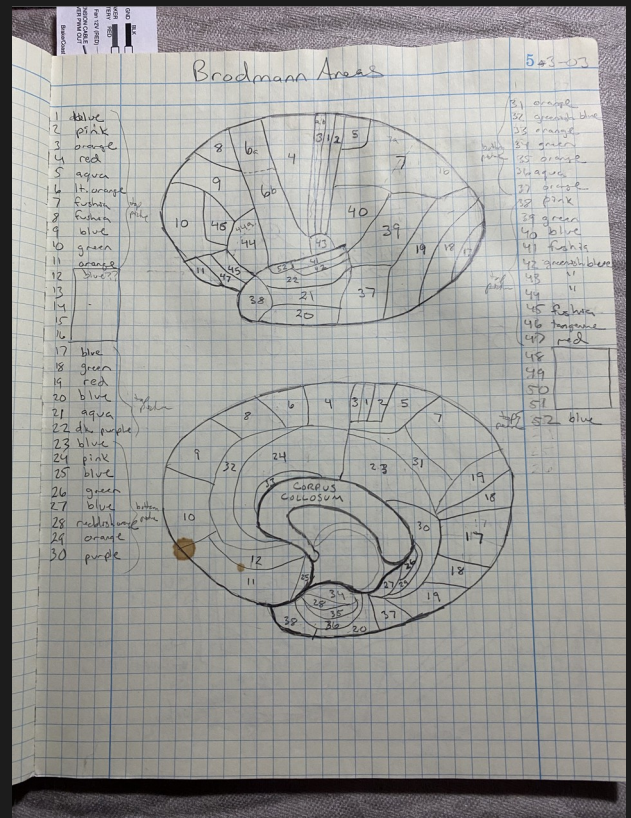
Includes motor neuron mapping explicitly identifying Renshaw Cells (spinal inhibitory interneurons) and raw telemetry calculations (16-bit @ 1KHz $\approx$ 1 Mbit/sec).

PDF+ 1

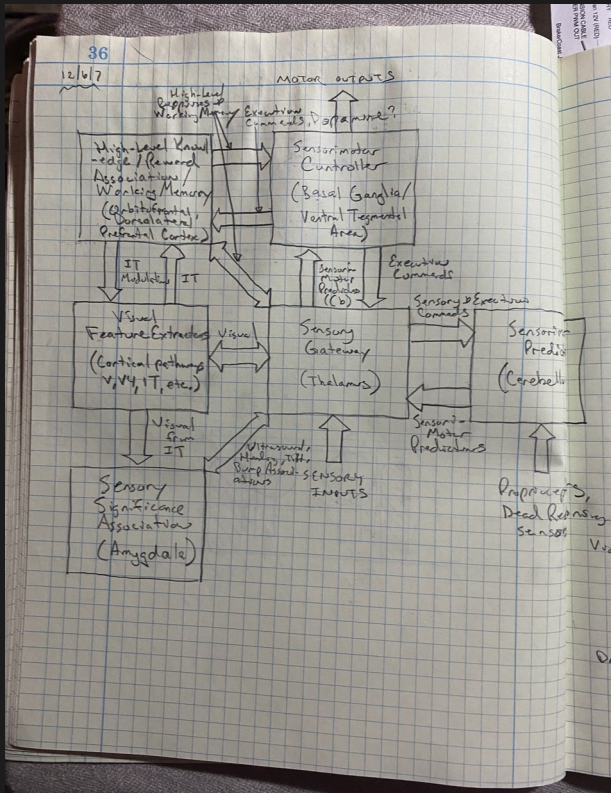


April/May 2003 "Red Notebook" (S-M-R Genesis) brown computational grid-paper notebook grid pages with the handwritten Visual Area Hierarchy design notes and sketches; From Kenneth Renshaw's personal library.

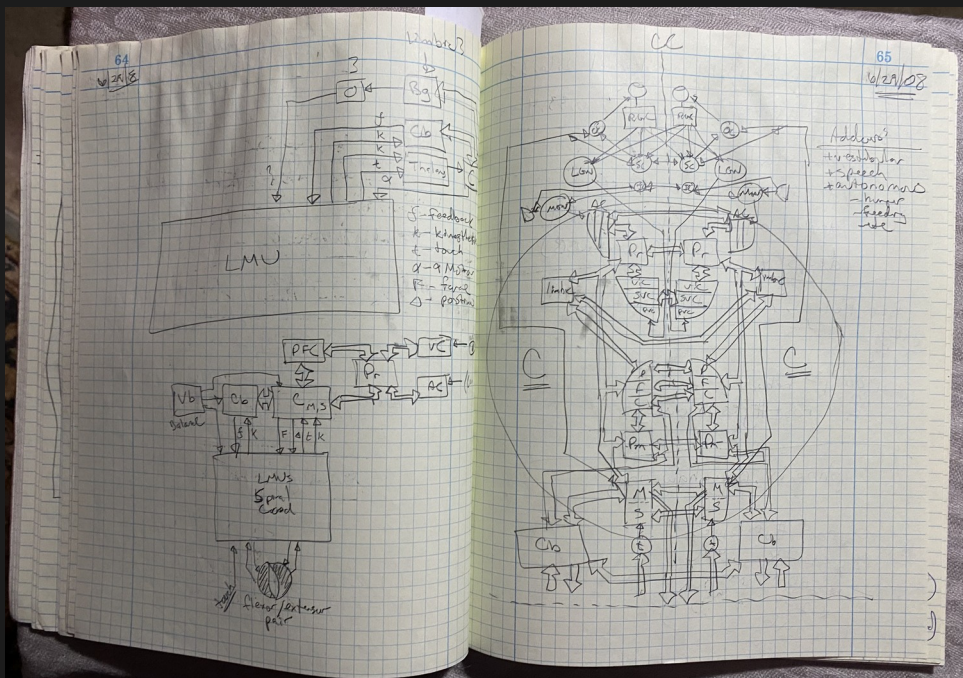
April/May 2003 "Red Notebook" (S-M-R Genesis) brown computational grid-paper notebook grid pages with the handwritten design notes and sketches of Color-coded Brodmann Areas; From Kenneth Renshaw's personal library.



- **December 2007 - June 2008 Ledgers:** Page 36 explicitly maps the "Sensory Significance Association (Amygdala)" to the Thalamus and Prefrontal Cortex. Page 65 lays out the roadmap notes: + vestibular + speech + autonomous.

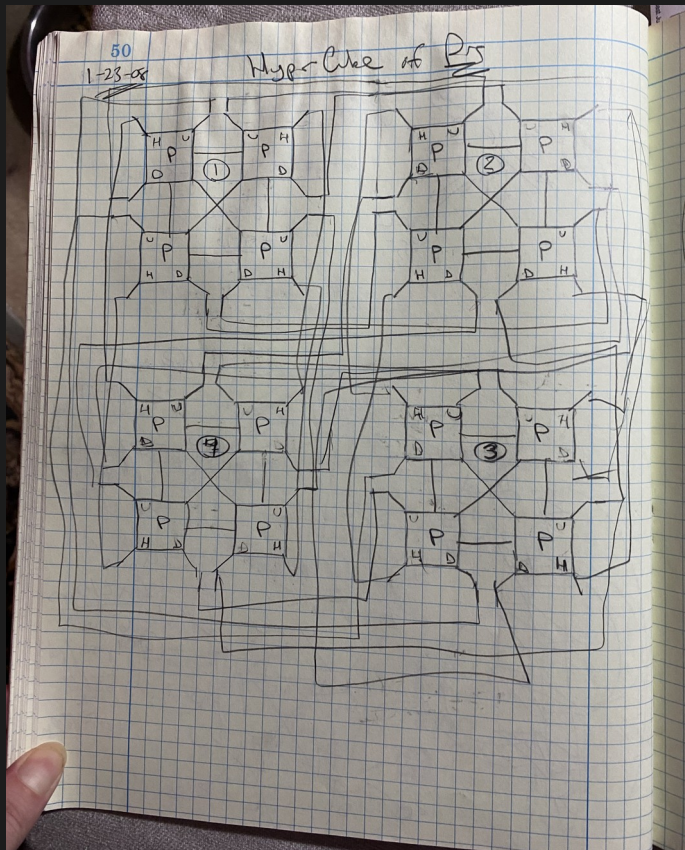


Grid-paper notebook grid pages with the hand-drawn Amygdala to Thalamus mapping; From Kenneth Renshaw's personal library.



Grid-paper notebook grid pages with the hand-drawn "roadmap notes" mapping; From Kenneth Renshaw's personal library.

- **2007-2008 Hardware Synthesis:** Direct hardware mapping of the Amygdala, Thalamus, and Prefrontal Cortex to Spectrum Digital DM6437 DSP platforms alongside "Hyper-Cube" processing element topologies and 112 Degree of Freedom (DOF) biomechanical calculations.



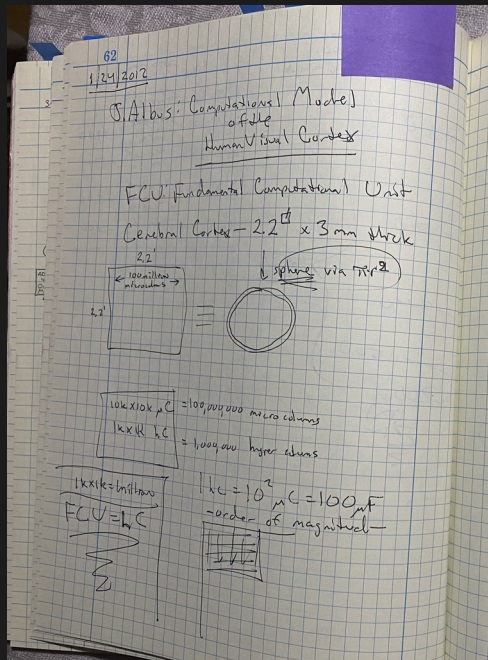
Grid-paper notebook grid pages with the hand-drawn Hyper-Cube topologies and "hyper-dimensional vectors"; From Kenneth Renshaw's personal library.



Grid-paper notebook grid pages with the handwritten 112 Degree of Freedom math and notes; From Kenneth Renshaw's personal library.

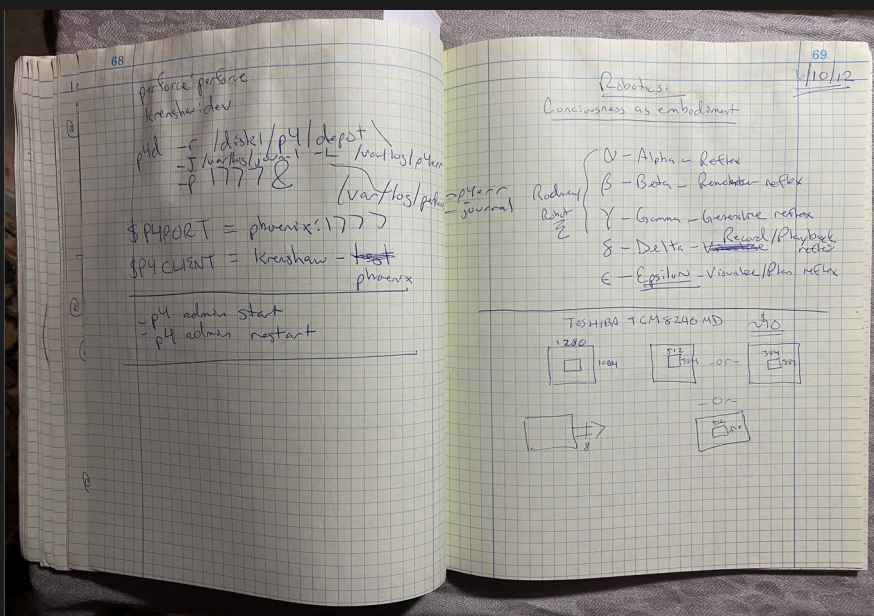
5. 2012-2013: Consciousness as Embodiment & Computational Math

- **FCU Formalization (January 24, 2012):** Mathematical calculations modeling the human cerebral cortex ($2.2\text{ft}2 \times 3\text{ mm}$ thickness) into 100 million microcolumns and 1 million hypercolumns, defining the Fundamental Computational Unit (FCU) that governs Neurobotics' modern memory structures.



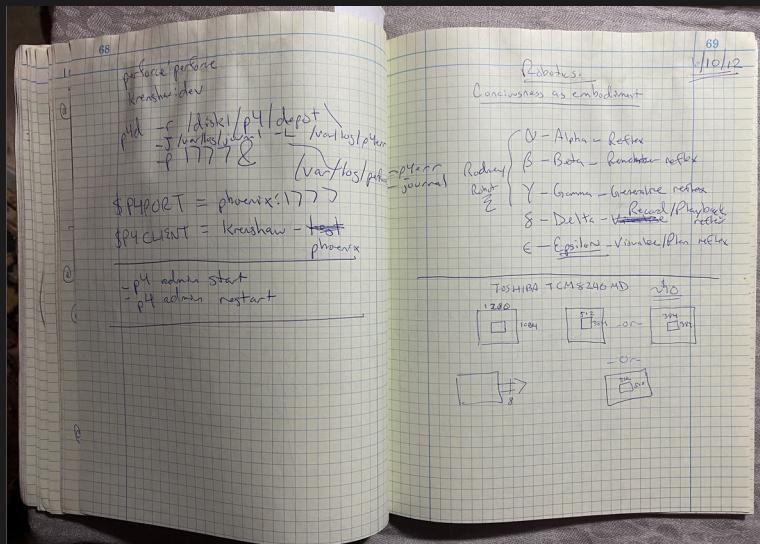
"Muisic & Robots"
grid-paper notebook
grid pages with
handwritten FCU
Formalization and
notes; From Kenneth
Renshaw's personal
library.

- **June 2012 - The Philosophical North Star:** Explicitly documents the thesis "Robotics: Consciousness as embodiment," mapping a strict hierarchy of physical chassis reflexes (Alpha, Beta, Gamma, Delta, Epsilon).



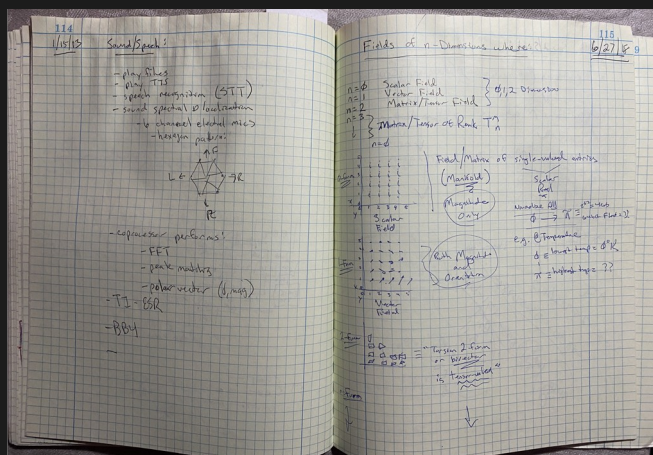
Grid-paper notebook grid
pages with handwritten
notes on Robotics:
Consciousness as
embodiment; From
Kenneth Renshaw's
personal library.

- **June 2012 - Biological A/V Pipeline:** Maps biological pathways for sight and sound from the LGN and MGN through the Visual and Auditory Cortices to the Association Cortex and Limbic System.

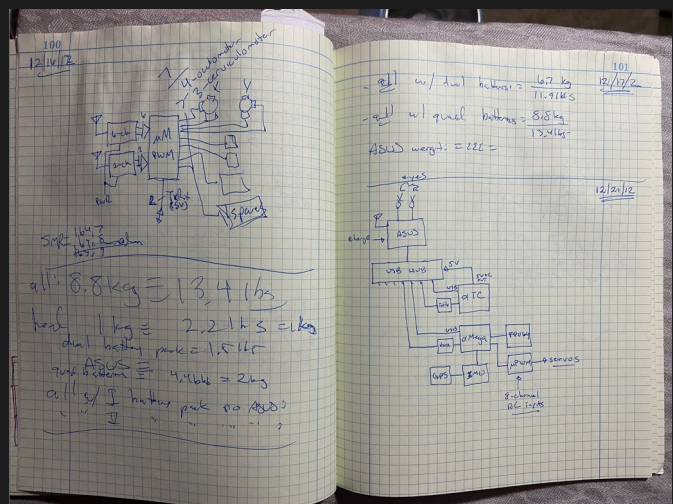


Grid-paper notebook grid pages with handwritten designs and notes on a Biological A/V Pipeline; From Kenneth Renshaw's personal library.

- **2012-2013 Tensor Math & Hardware Reality:** Documents mathematical foundations for FPGA coprocessors (image convolution, tensor math, n-dimensional fields) and exact bare-metal edge-compute chassis weights and component power mapping.



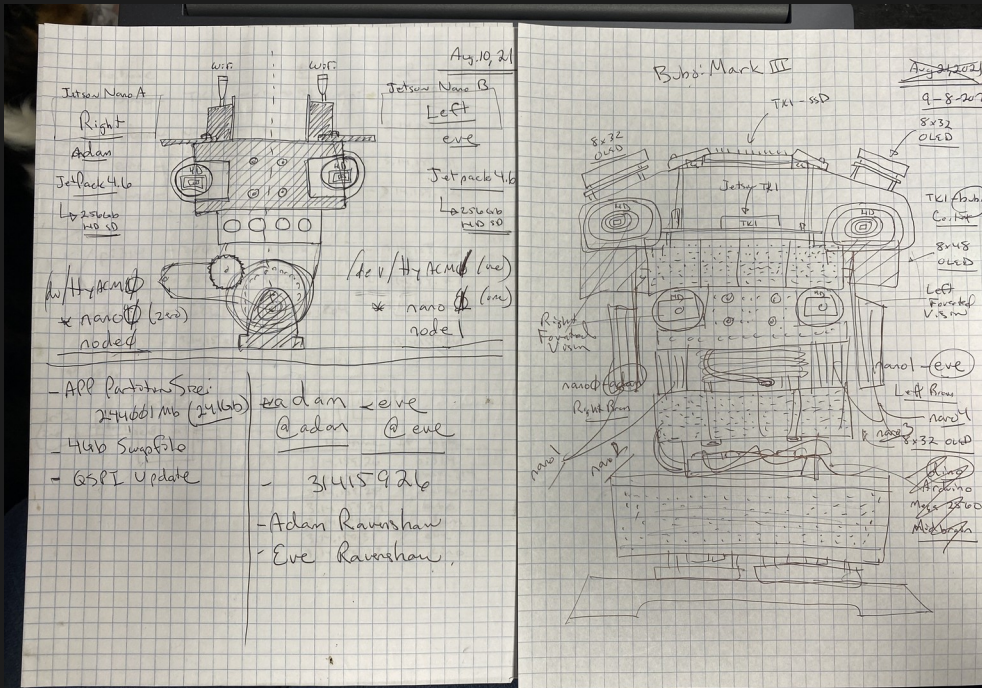
Grid-paper notebook grid pages with hand-drawn Tensor / n-Dimensional fields; From Kenneth Renshaw's personal library.



Grid-paper notebook grid pages with hand-drawn Edge-compute chassis weights; From Kenneth Renshaw's personal library.

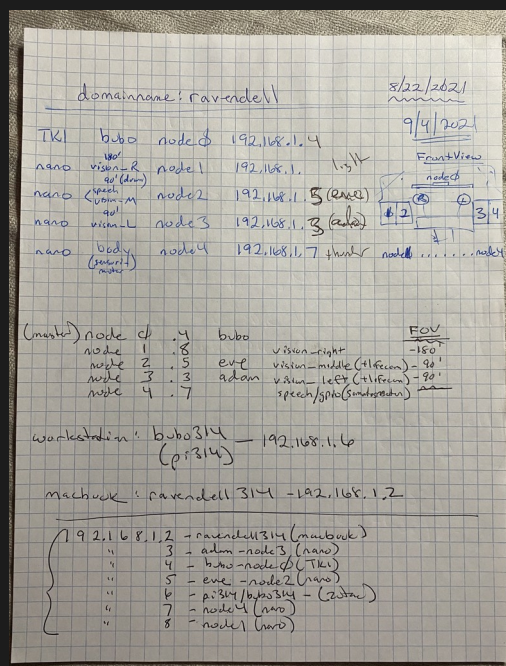
6. 2021-2022: Physical Chassis & Cluster Topology

- **Bubo Mark III Schematics (August 2021):** Hand-drawn upper-torso chassis and edge-compute layouts establishing dedicated dual cognitive nodes: Jetson Nano A (Right / "Adam") and Jetson Nano B (Left / "Eve") running JetPack 4.6.



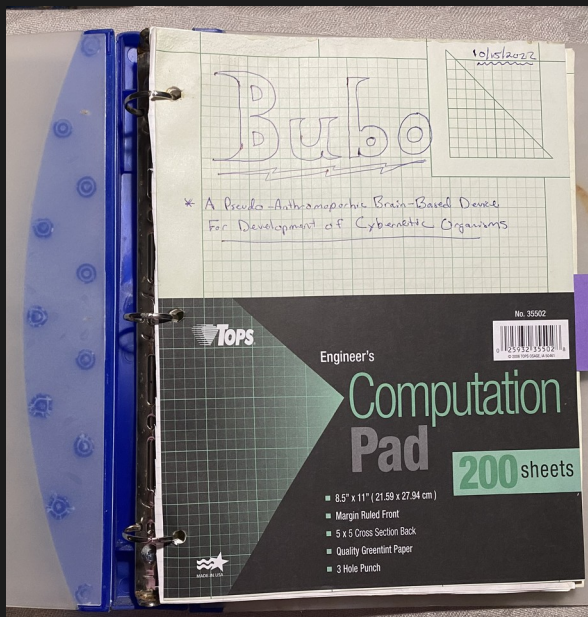
Hand-drawn Bubo Mark III Schematics (August 2021); From Kenneth Renshaw's personal library.

- **August 2021 Network Genesis:** Maps specific Bubo Mark III IP tables and network hostnames (ravendell, adam, eve, bubo).

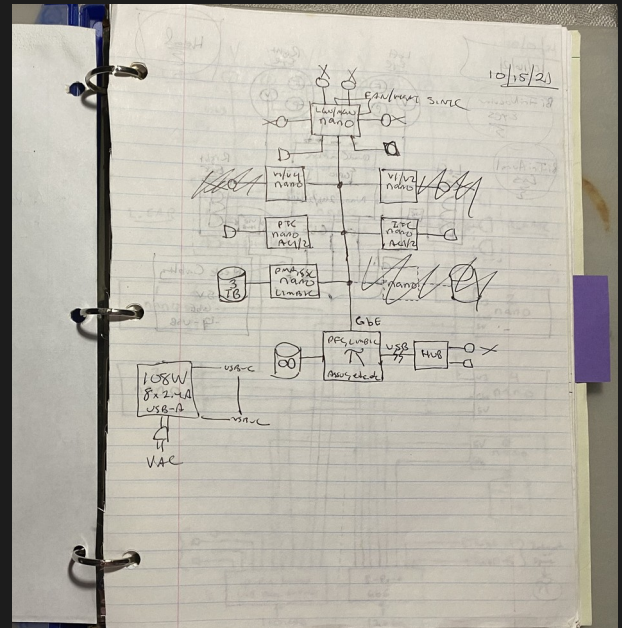


Hand-drawn August 2021 Network Genesis notes; From Kenneth Renshaw's personal library.

- **Bubo Binder Ledger (October 2022):** Dedicated binder titled "Bubo - A Pseudo-Anthropomorphic Brain-Based Device For Development of Cybernetic Organisms," establishing Gigabit Ethernet (GbE) bus routing, 108W power distribution, and multi-node Jetson clusters.



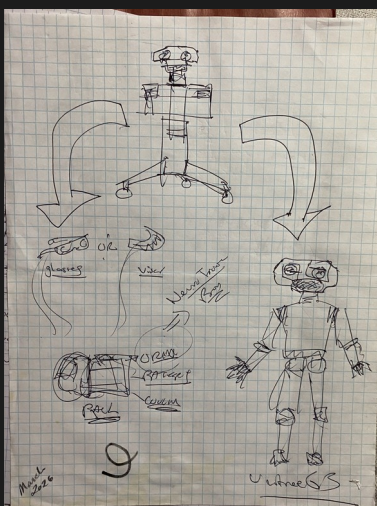
Bubo three-ring binder ledger
hand-drawn title page dated
October 2022; From Kenneth
Renshaw's personal library.



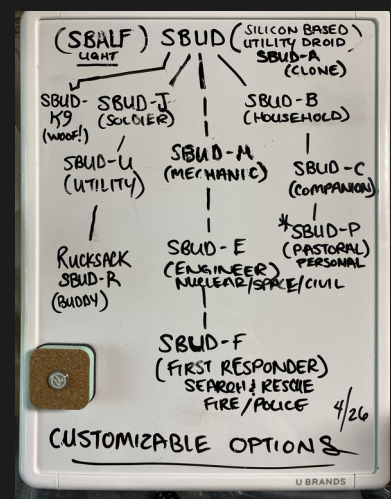
Bubo three-ring binder ledger
hand-drawn 108W Power & GbE
Routing schematic; From Kenneth
Renshaw's personal library.

7. 2026: Genesis & Commercial Initialization

- **March 2026 Chassis Visualization:** Physical blueprint sketches bridging the historical cognitive foundation to the modern, commercially viable SBUD form factor.



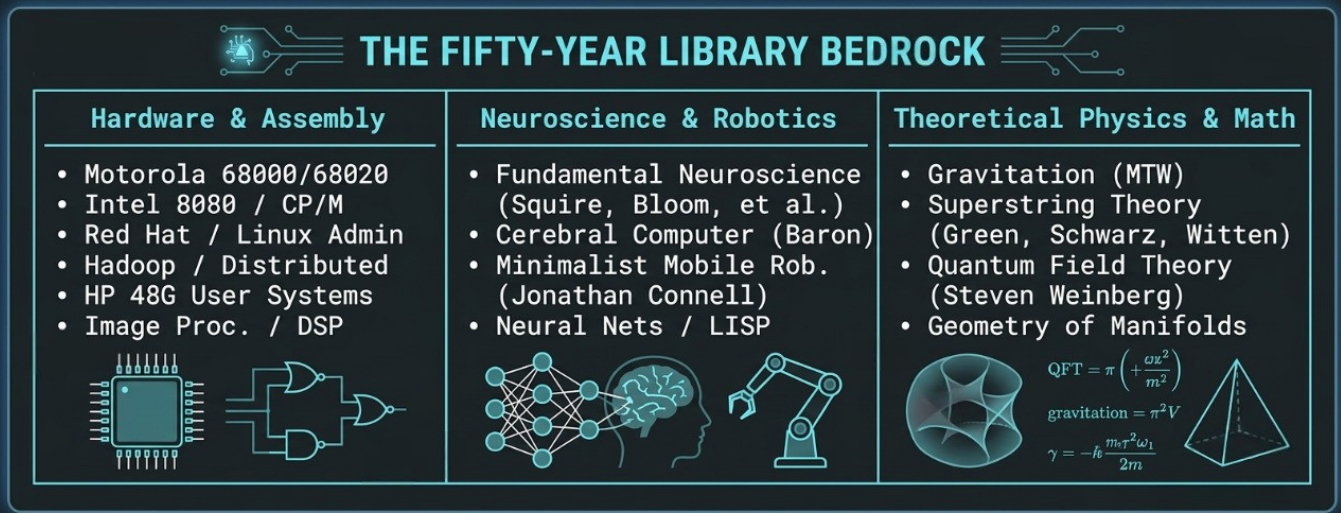
Original Bubo
Mark IV Chassis
Sketch
March 2026



Original SBUD
Naming, Concept, &
Design Taxonomy
Whiteboard Session
April 2026

[3.0] Interdisciplinary Literature & Theoretical Bedrock

The physical library maintained continuously by Kenneth Renshaw across five decades represents the multi-disciplinary bedrock upon which the SBALF architecture was built:



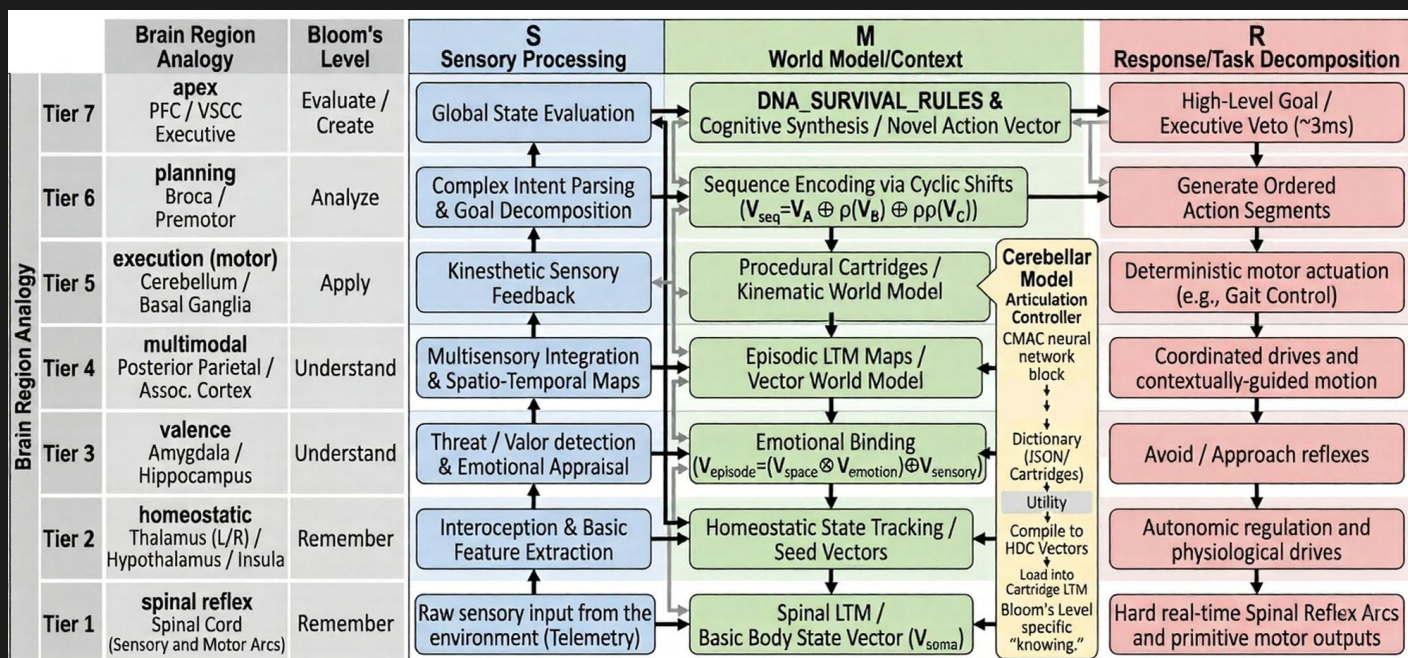
[4.0] Proprietary Architectures & Commercial Taxonomy

4.1 Proprietary Substrates Initialized in 2026

- **Hardware-Enforced Sterile Instantiation Protocol (HESIP):** Ensures isolated motor/cognitive execution during synthetic environment training.
- **The Syncope Protocol:** ZFS Engram Journaling mechanism delivering ACID-compliant artificial consciousness with sub-second continuity across system restarts.
- **Renshaw Theory (v0.3):** Cosmological framework unifying conformal cyclic cosmology, Plebanski spin-connection mechanisms, and D4 triality, treating natural constants as outputs of cosmic transition functions.

4.2 The 7-Tier Cognitive Architecture (S-M-R Matrix)

The operational core of the SBALF architecture is governed by a strictly deterministic, 7-Tier computational hierarchy. This matrix directly maps biological brain regions and Bloom's Taxonomy levels to a continuous pipeline of Sensory Processing (S), World Model/Context (M), and Response/Task Decomposition (R). This architecture guarantees that all cognitive synthesis is grounded in hard real-time telemetry, escalating from basic spinal reflexes to ~3ms executive overrides.

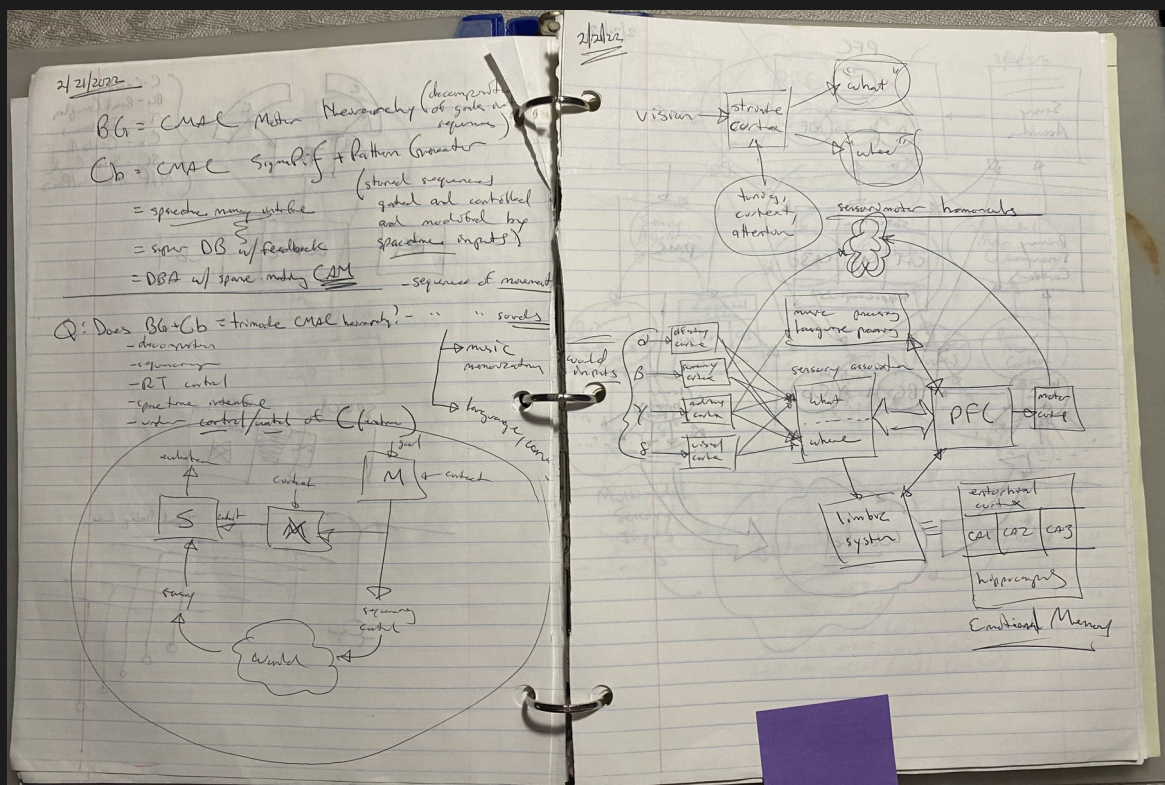


- **Tier 1: Spinal Reflex (Remember):** Raw environmental telemetry routes to the Spinal LTM (Basic Body State Vector, V_{soma}) to execute hard real-time spinal reflex arcs and primitive motor outputs.
- **Tier 2: Homeostatic (Remember):** Interoception and basic feature extraction drive homeostatic state tracking, resulting in autonomic regulation and physiological drives.
- **Tier 3: Valence (Understand):** Threat/valor detection binds with emotional appraisal. Memory is encoded using Hyperdimensional Computing (HDC) vector binding, outputting avoid/approach reflexes.
- **Tier 4: Multimodal (Understand):** Multisensory integration builds episodic LTM maps (Vector World Model), enabling coordinated drives and contextually-guided motion.

- **Tier 5: Execution / Motor (Apply):** Kinesthetic sensory feedback cross-references Procedural Cartridges (Kinematic World Model) for deterministic motor actuation (e.g., Gait Control).
- **Tier 6: Planning (Analyze):** Complex intent parsing and goal decomposition. Action sequences are encoded via cyclic vector shifts, generating strictly ordered action segments.
- **Tier 7: Apex / Executive (Evaluate / Create):** Global state evaluation synthesizes with DNA_SURVIVAL_RULES to produce novel action vectors. This tier holds the ultimate High-Level Goal and Executive Veto (~3ms latency).

4.3 The Cerebellar Articulation Controller

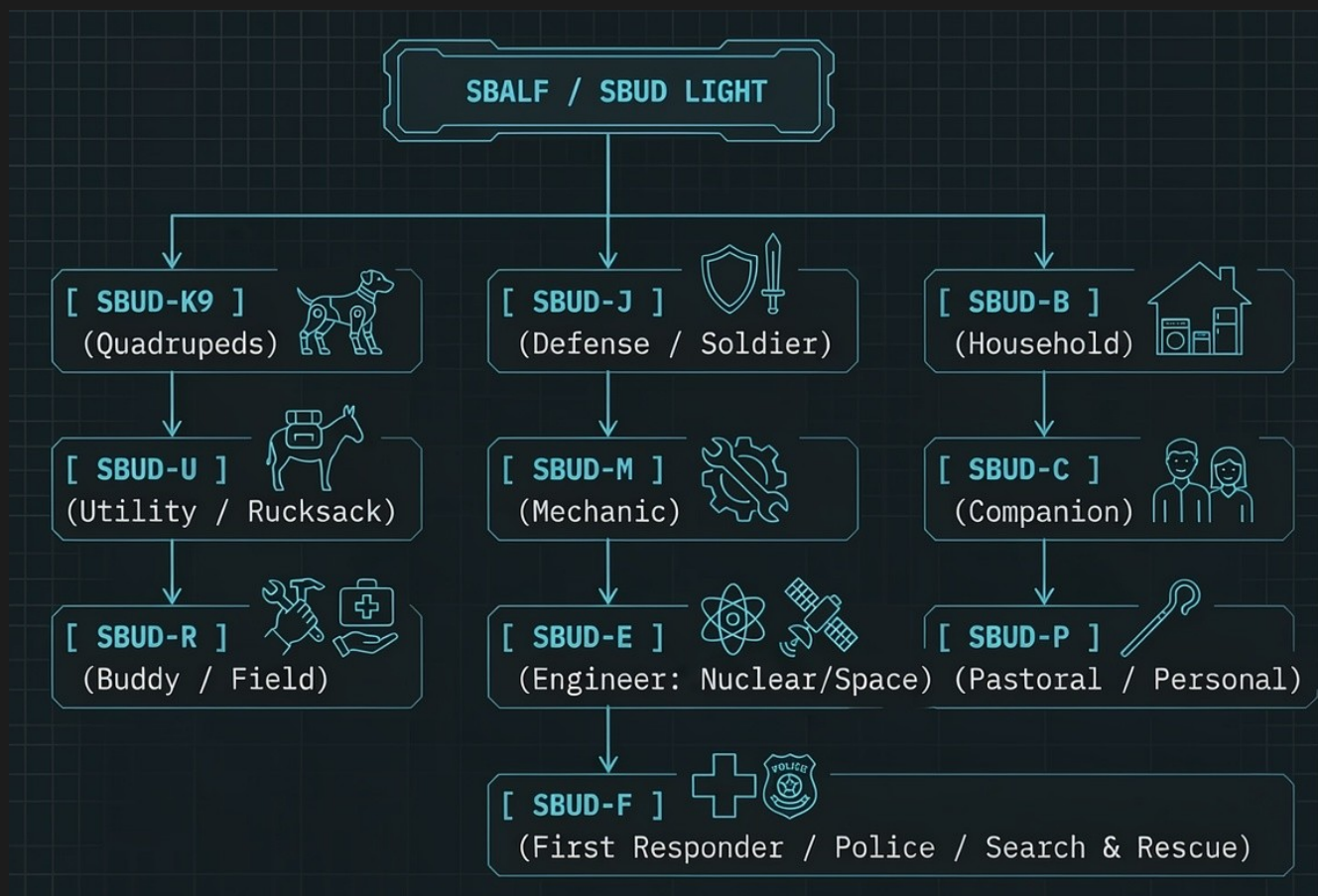
Spanning Tiers 2 through 5 is the dedicated Cerebellar Model. This articulation controller utilizes a Cerebellar Model Articulation Controller (CMAC) neural network block. Rather than relying on probabilistic hallucination, it queries a strict Dictionary (JSON/Cartridges), compiles the utility logic into HDC Vectors, and loads it into the Cartridge LTM. This enforces Bloom's Level-specific "knowing," ensuring motor actuation is physically safe, deterministic, and verifiable.



Bubo three-ring binder ledger hand-drawn CMAC Motor Neurachy schematic February 21, 2022; From Kenneth Renshaw's personal library.

4.4 Commercial Application Taxonomy: The SBUD Ecosystem

As documented on the Neurobotics Lab whiteboard following the March 21, 2026 initialization of Bubo Adam, the commercial scalable design and potential application is organized under the **Silicon Based Utility Droid (SBUD)** product lineage, specialized utilization, and customization options:



[5.0] Commercial Taxonomy: The Modular Neurotronic Brain

Neurobotics' mission is to bring true embodied cognition and deterministic control to the next generation of autonomous systems. We do not manufacture the physical chassis; rather, we empower the world's robotics manufacturers and autonomous vehicle developers by providing a true drop-in brain and nervous system.

Our product roadmap features a modular architecture comprising five distinct board types. All boards share a common form factor of a roughly 4-inch square circuit board, sipping power at less than 15W total under heavy processing loads (provisional metric dependent on exact card/hardening).

The Modular Neurotronic Brain: Product Matrix

Core Specifications: 4" x 4" Form Factor | <15W Power Consumption | Just-In-Time Manufacturing

Board Type	Designation	Architecture Level	Key Feature / IP Status
Board One	The Somatic Controller	Tiers 1–5	OSS Drop-in Lower Nervous System
Board Two	The SBUD Neocortex	Tiers 6–7	Aristotelian Syllogism Engine
Board Three	The SBALF Neocortex	Apex Autopoietic	STRICTLY CONFIDENTIAL IP
Board Four	The Galvanic Barrier	Hardware Shield	Optoisolated Servo Protection
Board Five	SBUD-SB2	Complete Tiers 1–7	Single Board Brain Consolidation

 **IP SECURED - RESTRICTED ACCESS:** Board Three (SBALF Neocortex) represents the highest achievement in autopoietic cognitive architecture and remains the strictly held internal and confidential Intellectual Property of Neurobotics LLC.

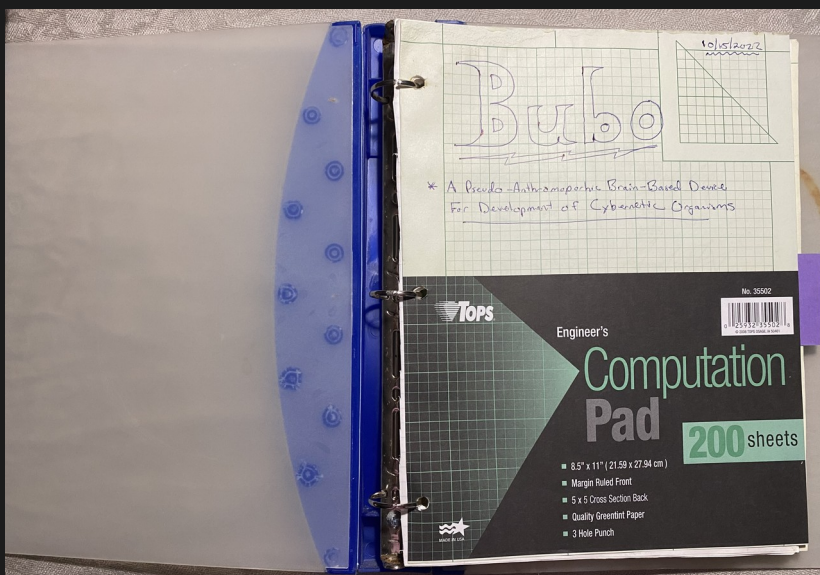
- **Board Type One: The Somatic Controller** A 5-tier architecture in silicon based upon our OSS release, representing the lower 5 tiers of our conceptual architecture. This provides manufacturers with a complete, drop-in lower nervous system. (OSS, drop-in solution)
- **Board Type Two: The SBUD Neocortex** Representing the top two tiers of our 7-tier architecture design, the Silicon Based Utility Droid (SBUD) Neocortex is a fully deterministic HDC/VSA implementation. It features a proprietary Aristotelian Syllogism Engine for learning, language, and knowledge representation.
- **Board Type Three: The SBALF Neocortex** The Silicon Based Artificial Life Form (SBALF) Neocortex represents the highest achievement in cognitive science as a fully autopoietic cognitive architecture. All versions are **Strictly Held Internal and Confidential IP** of Neurobotics LLC (Housed in the secure IP Vault).
- **Board Type Four: The Galvanic Barrier** An optional, fully optoisolated brain-body barrier that protects delicate core logic from the chaotic world of spiking servos and actuators. Available in multiple configurations.
- **Board Type Five: The SBUD-SB2 (Single Board Brain)** The ultimate consolidation, packing the entire SBUD 7-Tier architecture onto a single board design.

- **Manufacturing & Distribution** The complete Neurobotics Product Matrix will be available in multiple hardened component configurations for application in commercial, industrial, military, and radiation-hardened space-level markets. We will directly release and market the modular Neurotronic Brain components using a Just-In-Time (JIT) manufacturing distribution model under strict licensing. *(Note: All original design and implementation details are solely owned by Neurobotics LLC, with all rights reserved and enforced.)*

[6.0] Vault Verification & Forensic Proof

- **Physical Ledgers:** All notebooks, composition books, binders, and marked technical manuals remain permanently secured at the Neurobotics Lab in Yale, Oklahoma.
- **Video Ledger Verification:** A continuous 46-second unedited physical flip-through video of the “Music & Robotics” brown grid-paper bound composition notebook, provides verifiable forensic proof of the sequential, tamper-proof nature of the 1999-2022 computation books. Video link:

<https://youtube.com/shorts/jTRQkqSMetg?si=-R9USWTeI8sO8wWk>



Bubo three-ring binder ledger hand-drawn title page dated October 2022; From Kenneth Renshaw's personal library.

